DOCTORAL CONSORTIUM · ASSETS 2026

*Designing **Autonomy-Preserving** Cognitive Support*

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Why autonomy matters

Some people need more support with memory, attention and planning. Neurodivergent children develop these skills differently. Older adults' skills change with age.

Technology can give this support at scale. But it is mostly built to finish the task, so it does the task for people and not with them. This can undermine their sense of autonomy, a basic psychological need.

Helping
A nudge toward the right piece.

Doing for
Placing the piece for you.

Three considerations recur in related work

1 How much
Fits what the person can still do.

2 How
Collaborative, not imposed.

3 For whom
Their own goals, not assumptions.

An adaptive robot acts on an estimate of what the person needs, so adapting to the person carries a paternalism risk.

To our knowledge, no published work has deployed a robot for cognitive support with autonomy as an explicit goal.

[RQ] Robot as testbed

How should **cognitive support be designed** to preserve users' sense of **autonomy**, and how can **assistive robots** contribute?

RQ1 Estimate cognitive state from what users do
Expected: behavioral indicators checked against standardized measures, and a reusable validation method.
Work completed

RQ2 Design robot behaviors that preserve autonomy
Expected: a repertoire of autonomy-preserving robot behaviors.
Work completed

RQ3 Adapt when and how the robot supports
Expected: a robot that adapts without an operator, and evidence on its effect on autonomy and performance.
Work remaining

Study 1 · Gameplay

Under review, CHI 2027

81 neurodivergent children, ages 7 to 14



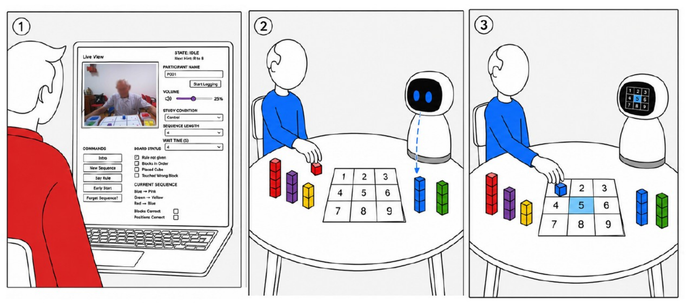
Games by the EMPOWER consortium.

- Four games adapted from standardized tasks.
- Three of four aligned with their standardized task at some levels. Sustained attention hit a ceiling.
- Features beyond the score were meaningful.
- The main contribution is a reusable validation method.

Study 2 · Robot hints

Under review, HRI 2027

35 older adults, Wizard-of-Oz memory task



Perceived autonomy (single item)

4.61 Implicit	1.82 Explicit

- A robot gave hints, from subtle to direct.
- Implicit hints gave higher perceived autonomy, with no cost to performance. They needed more hints.
- Early evidence. The sample was small.

Next steps

The final study is not decided. Options:

- **Adaptive robot.** An adaptive policy against fixed implicit support.
- **Offering choice.** Buttons for "help" and "let me try".
- **Assumption gap.** Do designers and carers predict what older adults prefer?
- **Robot or screen.** Does the robot add anything?

The task could be groceries or medication management.

Nov to Jan	Study design, ethics
Jan to Mar	Implementation, pilots
Apr to Jun	Data collection
Jul to Sep	Analysis, paper
Sep to Dec 2027	Thesis writing

Limitation. No study yet combines assessment and robot support in the same population and task.

Questions for you

1 Which final study would you run?

2 How would you measure autonomy with older adults?

3 Is giving people control over help the right direction?

4 Does this story hold together for accessibility research?

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